EE3032 Introduction to VLSI Design—Homework 3

1. Fig. 1 shows a two-input AND gate. Assume that all the transistors have the same size. Simulate your design using Hspice simulator and measure the worst-case high-to-low and low-to-high propagation delays at node Z and F. (5 points)

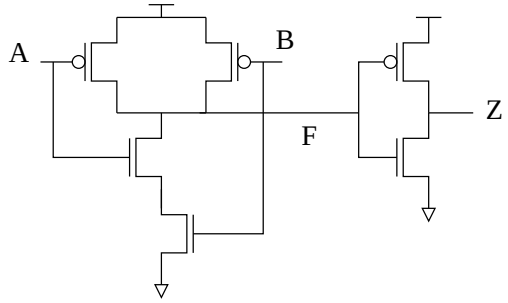

Figure 1: An AND gate.

2. The dynamic power consumption (P_d) of a CMOS circuit is composed of switching power consumption P_{sw} and short-circuit power consumption (P_{sc}). Consider an inverter driving a load C_L shown in Fig. 2. Complete the curve of P_{sc} associated with C_L, and the value of C_L is ranged from 0.1pf, 0.3pf, 0.5pdf, 0.7pf to 1pf using Hspice. Check if the P_{sc} is associated with the C_L. If yes, why? (5 points)

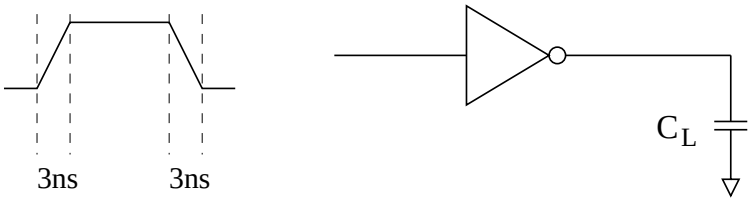

Figure 2: Inverter.